Analyzing Lexical Variance and Morphological Collapse in LLM-Generated vs. Human-Written Marathi Text via PCA

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Abstract

Generative Large Language Models (LLMs) often struggle to replicate the complex morphological variations natural to highly inflectional languages like Marathi. This paper investigates whether Principal Components Analysis (PCA) can effectively separate human-written Marathi from machine-generated Marathi by reducing a high-dimensional matrix of lexical and morphological features. We propose an interpretable pipeline to differentiate LLM-generated and human-written text. Hand-crafted features, particularly capturing Lexical Diversity and Morphological Variance, are extracted and reduced via PCA, capturing 40.15% variance across the first two principal components. A Support Vector Machine (SVM) classifier trained on these components achieves 92.29% accuracy, significantly outperforming a baseline BERT embeddings approach coupled with PCA (56.91%). Our analysis reveals that lexical diversity and morphological depth are strong discriminators, highlighting the "morphological collapse" in current LLMs during Marathi text generation. Code is publically available at https://github.com/Maitreya152/LD3_Final_Project

1 Introduction

The rapid advancement of Generative Large Language Models (LLMs) has led to highly fluent text generation in many languages. However, generating text in low-resource, morphologically rich languages like Marathi (an 8^{th} Schedule language of India) remains a challenge. Current open-source models often default to statistically rigid structures, failing to replicate the nuanced morphological inflections naturally employed by human writers. This phenomenon can be characterized as "morphological collapse."

Detecting machine-generated text in such languages poses a dual utility: it allows us to audit the capabilities of LLMs and provides a reliable mechanism to distinguish artificial content. Much of the recent work in machine-generated text detection relies on large neural embeddings, which act as black boxes and offer little insight into the specific linguistic mechanisms that LLMs fail to capture.

In this work, we pose the following research questions:

- **RQ1:** Can Principal Components Analysis (PCA) effectively separate human-written Marathi from machine-generated Marathi by reducing a high-dimensional matrix of handcrafted lexical and morphological features?
- **RQ2:** Which specific linguistic features (e.g., morphological complexity, prefix/postfix variance, or top-k vocabulary overlap) dominate the principal components that serve as the axes of separation between the two text sources?
- **RQ3:** How does an interpretable pipeline based on handcrafted (HC) features compare against a dense baseline using pre-trained multilingual contextual embeddings for the task of classifying Marathi text origin?

2 Related Work

2.1 Machine-Generated Text Detection

The challenge of detecting machine-generated text has seen a surge of interest alongside the rapid deployment of powerful LLMs (Jawahar et al., 2020), (Bakhtin et al., 2019). Early approaches, such as the GLTR (Giant Language Model Test Room) framework (Gehrmann et al., 2019), demonstrate that machine-generated text often heavily samples from the most predictable words in a model's vocabulary, creating an unnatural distribution of token probabilities. While predictability-based metrics were effective for English text generated by earlier models, extending them to highly inflectional languages like Marathi requires further investigation. Our research addresses this gap.

Another study by Kendro et al. (2026) investigates the patterns of lexical diversity in LLM-generated texts by GPT models in comparison with texts written by L1 and L2 English participants various education levels. Six dimensions of lexical diversity were measured in each text: volume, abundance, variety-repetition, evenness, disparity, and dispersion. Results from Support Vector Machines (SVMs) (Moguerza and Muñoz, 2006) reveal that the LLM-generated texts differ significantly from human-written texts with LLMs showing higher levels of lexical diversity despite producing fewer tokens. On the other hand, human writer's lexical diversity did not differ across subgroups like education, language status etc. Joseph et al. (2025) discusses various other stylometric features, lexical richness, syntactic complexity, function word frequency, and character-level patterns, to train classifiers (e.g., SVM, Random Forest) which show clear stylistic differences between the LLM-generated and human-written text. Following these approaches, we evaluate lexical and syntactic features in the context of Marathi. In addition to standard classifiers, we employ PCA (Shlens, 2014) to analyze the underlying variance between human and LLM-generated Marathi text.

2.2 Marathi NLP and Morphological Challenges

Marathi is an Indo-Aryan language characterized by a highly agglutinative and inflectional morphology. Nouns, verbs, and adjectives undergo complex transformations depending on gender, number, case, and tense (Dhongde and Wali, 2009). For instance, a single root word can take dozens of affixed forms. Most open-source LLMs employ subword tokenization strategies (e.g., Byte-Pair Encoding or WordPiece) heavily optimized for English. When applied to languages like Marathi, these tokenizers often fragment words unnaturally, causing models to struggle with constructing grammatically perfect and diverse inflections. While resources like the L3Cube-MahaNews corpus (Mittal et al., 2023) provide a standard baseline of high-quality human-written news articles, there remains a significant gap in evaluating the structural and morphological divergence between such human text and text generated by modern LLMs. Our work bridges this gap by explicitly calculating and using morphological variance metrics.

3 Dataset

To answer our research questions, we constructed a comparative dataset consisting of two parallel sets:

- **Human Text Baseline:** A sample of 2,700 human-written Marathi news articles derived from the standard L3Cube-MahaNews corpus (Mittal et al., 2023).
- **LLM-Generated Text:** 2,700 Marathi articles generated by an open-source model (*Llama-3.2-3B* (Grattafiori et al., 2024)). The model was prompted with the exact headlines corresponding to the human-written news dataset to ensure thematic consistency.

4 Methodology

Inspired by the statistical analysis of GLTR (Gehrmann et al., 2019), our pipeline utilizes an unsupervised dimensionality reduction technique to expose differences in the underlying feature distributions.

4.1 Feature Extraction

To encapsulate the linguistic nuances of Marathi, we utilized the IndicNLP library (Kunchukuttan, 2020) for robust text tokenization and Stanza (Qi et al., 2020) for deep dependency parsing. We extracted a comprehensive set of document-level features categorized into four distinct linguistic groups, designed to capture both superficial statistics and deeper structural patterns:

1. **Lexical Features (3):**
    - *Type-Token Ratio (TTR):* Quantifies vocabulary diversity.
    - *Hapax Legomena Ratio:* The proportion of words appearing exactly once in the document.
    - *Average Word Length:* Mean number of characters per word.
2. **Syntactic Features — Basic (3):**
    - *Average Sentence Length:* The mean number of words per sentence.
    - *Sentence Length Variance:* Evaluates the variation in sentence lengths.
    - *Punctuation Density:* The ratio of punctuation marks to words.

3. **Predictability Features (2):** Drawing direct inspiration from GLTR (Gehrmann et al., 2019), we approximate text predictability using 2 features:
   - *Top-100 Vocab Overlap %:* The fraction of tokens belonging to the 100 most frequent words in a reference corpus.
   - *Top-1000 Vocab Overlap %:* The fraction of tokens belonging to the 1000 most frequent words.

4. **Morphological Feature (1):**
   - *Morphological Complexity:* Estimated by the average number of subword tokens produced per word by a standard BPE tokenizer.

5. **Syntactic Features — Deep (4):**
   - *Average Dependency Length:* Absolute linear distance between head and dependent tokens in sentences.
   - *Unique POS Ratio:* The diversity of POS tags relative to text length.
   - *Prefix Variance:* Computed by isolating initial character n-grams.
   - *Postfix Variance:* Computed by isolating terminal character n-grams across tokens (highly relevant for Marathi agglutination).

6. **Structural Features (4):**
   - *POS Bigram Entropy:* Lower values indicate reliance on highly repetitive, rigid grammatical sequences.
   - *Dependency Pattern Diversity:* Variety of syntactic root connections.
   - *Average Tree Depth:* Mean depth of the parsed dependency tree.
   - *SOV Order Ratio:* The frequency at which clauses adhere strictly to the traditional Subject-Object-Verb construction.

7. **POS Density Features (5):**
   - *Noun Ratio:* The proportion of tokens tagged as nouns.
   - *Verb Ratio:* The proportion of tokens tagged as verbs.
   - *Adjective Ratio:* The proportion of tokens tagged as adjectives.
   - *Adverb Ratio:* The proportion of tokens tagged as adverbs.
   - *Proper Noun Ratio:* The proportion of tokens tagged specifically as proper nouns.

4.2 Standardization and PCA

Because our engineered features exist on vastly different numerical scales (e.g., ratios bounded between 0 and 1 versus tree depths averaging around 5), standardization is imperative. We apply standard scaling ($z = (x - \mu)/\sigma$) to center the mean at zero and scale the variance to one for each feature.

Subsequently, we apply Principal Components Analysis (PCA) to the standardized matrix. PCA computes the eigenvectors and eigenvalues of the data's covariance matrix, allowing us to extract the top orthogonal principal components. These components represent the axes of maximum variance in our high-dimensional linguistic space, facilitating an unsupervised visual separation of the data.

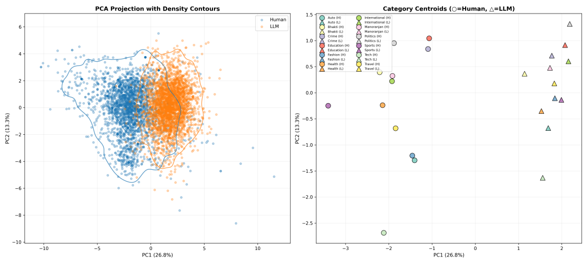

Figure 1: PCA Scatter Plot showing strong separability between Human and LLM-generated Marathi text across the first two principal components.

4.3 Classification Formulation

We formulate text source detection as a downstream classification task. We project the documents into the reduced 2D PCA space and train a Support Vector Machine (SVM) to classify the origin of the text. Analyzing the component loadings allows for robust interpretability regarding which features contribute most to the classification.

To evaluate effectiveness, we perform comparative benchmarking against a baseline using raw BERT (Devlin et al., 2019) embeddings reduced via PCA, providing a direct contrast between handcrafted explicit features versus dense representations.

5 Results and Key Findings

Our evaluation focuses on the interpretability of PCA components and the downstream classification performance.

5.1 Principal Component Loadings (RQ2)

The PCA reveals strong linear separability within the first two components, which together explain 40.15% of the total variance:

- **PC1 (26.80% Variance):** The primary axis of separation is heavily dominated by *Lexical Diversity* (Type-Token Ratio, Hapax Ratio) and *Morphological Variance* (Prefix/Postfix Variance, Tree Depth). A shift towards the positive PC1 direction strongly correlates with higher lexical diversity, characterizing human-written text.

- **PC2 (13.35% Variance):** The secondary axis captures *POS Composition* (Noun/Verb ratios) and *Syntactic Regularity*. Notably, POS Bigram Entropy exhibits a strong negative loading, confirming that LLM-generated Marathi text sequences fall back onto highly predictable and rigid part-of-speech patterns.

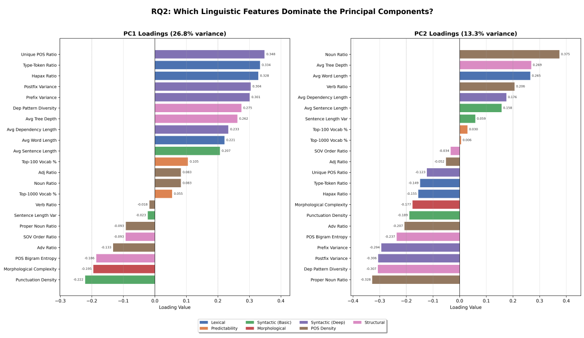

Figure 2: Feature Loadings for PC1 and PC2. Lexical diversity and morphological features strongly drive PC1.

Interestingly, the predictability features inspired by GLTR (Top-K Vocabulary Overlap) exhibited very weak loadings on both principal components. This is a critical finding: it suggests that for highly inflectional languages like Marathi, deep morphological and structural variance features are significantly stronger discriminators of text origin than pure vocabulary frequency overlaps which dominate English text detection.

Answering RQ2: Our component loading analysis explicitly demonstrates that lexical diversity metrics (like TTR and Hapax Ratio) alongside deep structural morphological features (such as Prefix/Postfix Variance and Tree Depth) dominate the principal components separating the two text sources. In highly inflectional languages, morphological complexity serves as the primary axis of separation between naturally diverse human writing and rigid LLM generation.

5.2 Classification Performance versus Baseline (RQ1 & RQ3)

To answer RQ1 and formally benchmark our approach against modern dense architectures (RQ3), we project our documents into the 2D PCA space and train a Support Vector Machine (SVM) equipped with an RBF kernel. We simultaneously evaluate a baseline utilizing average-pooled *bert-base-multilingual-cased* embeddings.

Table 1 summarizes the classification accuracies achieved across our five experimental pipelines.

Pipeline Architecture	**Accuracy**
HC Features → PCA → SVM	**0.9229**
HC Features → SVM (No PCA)	0.9558
BERT Embeds → PCA → SVM	0.5691
BERT Embeds → SVM (No PCA)	**0.9690**
HC + BERT → PCA → SVM	0.6754

Table 1: Classification Performance across Pipelines. HC implies Handcrafted features.

Answering RQ1: The handcrafted linguistic features processed through the unsupervised PCA reduction and classified by an SVM yield an impressive accuracy of 92.29%. This conclusively proves that PCA effectively discovers the linear combinations of morphological and lexical features necessary to distinguish between machine and human Marathi text.

Answering RQ3: When comparing our methodology to the dense contextual baseline, the HC Features with PCA pipeline dramatically outperforms the BERT embeddings subjected to the same PCA reduction (92.29% vs. 56.91%). When raw features are fed directly into an SVM without dimensionality reduction, BERT achieves a marginally higher peak accuracy (96.90%) compared to our explicit HC formulation (95.58%).

However, this marginal accuracy gain of 1.3% comes at the complete cost of interpretability. The utter collapse of BERT's performance when bottlenecked through PCA (56.91%) underscores the entangled, diffuse nature of transformer-based latent representations. In contrast, our handcrafted features maintain highly orthogonal variance mapping directly to human-understandable linguistic concepts. The results suggest that for auditing LLM text generation quality, simpler, linguistically-grounded features are highly competitive with dense embeddings, while providing indispensable analytical transparency.

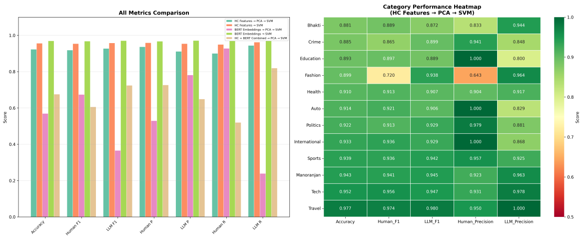

Figure 3: Detailed performance comparison across different features and classification methods.

5.3 Qualitative Analysis

To ground our quantitative findings, we present representative qualitative examples from the dataset that illustrate the structural and morphological discrepancies isolated by our PCA.

Example 1: Human-Written Marathi (High Lexical Diversity) *Text:* भारतासह जगभरातील कंझ्युमर इलेक्ट्रॉनिक्स आणि ऑटो कंपन्यांना सध्या सेमीकंडक्टर चिप आणि कॉम्पोनन्ट्सच्या तीव्र टंचाईचा सामना करावा लागत आहे. पुरवठा साखळीतील अडथळ्यांमुळे वाहतूक खर्चातही सहापट वाढ झाली आहे.
Analysis: This excerpt demonstrates high lexical diversity (Type-Token Ratio $\approx$ 0.97) and deep morphological complexity. The human author correctly synthesizes heavily agglutinized domain-specific vocabulary (e.g., कॉम्पोनन्ट्सच्या *[-cyā/of]*, *[-nmulē/due to]*), utilizing varied postfixes that directly contribute to the high positive PCA loading on *Prefix/Postfix Variance*.

Example 2: LLM-Generated Marathi (Rigid Syntactic Structure) *Text:* मुंबईतील त्याळे शहरातील प्रसिद्ध वाहन निर्माता कंपनी बोलो टू व्हीकल्स लिमिटेडच्या कारखान्यावर हालचाल झाली. या कंपनीने सध्याच्या दिवशी आपल्या नवीन उत्पादन मॉडेल 'अल्टो'वर आधुनिक कारखान्यांसह-सामंजस्यी विनियमित करण्यासाठी अंतर्गत परिशिष्ट जाहीर केले.
Analysis: While grammatically passable, the LLM hallucinates facts (e.g., बोलो टू व्हीकल्स लिमिटेड) and relies on highly rigid and repetitive POS sequences. The text exhibits a lower Type-Token Ratio ($\approx$ 0.84) and shallower dependency trees, mapping to the negative region of our principal components associated with "morphological collapse."

6 Conclusion

This paper demonstrated that PCA applied to handcrafted linguistic features can effectively and interpretably separate machine-generated Marathi content from human-written text. By defining an explicit set of morphological, lexical, syntactic, and predictability metrics, our pipeline achieved an accuracy of 92.29% in the reduced 2D PCA space. This performance heavily outperformed identically reduced dense BERT embeddings. Unlike deep continuous models which obscure the mechanisms of generation, our approach provides explicit insights into the functional shortcomings of current LLMs.

Specifically, we identified that LLM-generated Marathi suffers drastically from "morphological collapse." Comparing human text to machine generations, LLMs exhibit statistically significant drops in lexical diversity, possess much lower prefix and postfix variance, output shallower dependency trees, and demonstrate unnaturally regular and predictable POS bigram entropy. In essence, current generative models default to statistically rigid, grammatically safe structures rather than electing the naturally diverse and complex agglutinations inherent to native human Marathi writers.

Future Work

Future research will expand this methodology to other Indic languages, such as Hindi, Bengali, and Tamil, to test whether these morphological features generalize across different language families. Additionally, we plan to evaluate newer open-source and proprietary LLMs (e.g., GPT-5, Gemini-3) to test the robustness of our detection pipeline as inherent model generation quality improves. Methodologically, exploring the inclusion of more PCA components (e.g., 5, 10, or 20) will help identify the optimal dimensionality-accuracy tradeoff for classification. We also aim to conduct rigorous cross-domain evaluations—training the SVM on one text category and testing on others—to validate that our handcrafted features capture true source-level stylistic patterns rather than mere topic-level artifacts. Finally, determining the minimal optimal feature subset through advanced feature selection techniques, such as LASSO regression or mutual information scoring, remains a key objective for streamlining real-time detection systems.

Limitations

The primary limitation of our empirical study is its reliance on a single open-source LLM (*Llama-3.2-3B*) as the baseline generation source. Different parameter scales and proprietary architectures (e.g., GPT-4o or Claude 3.5) may exhibit varying degrees of morphological collapse when prompted in Marathi. Furthermore, since both our synthetic and human-written texts are restricted to the news domain (*L3Cube-MahaNews*), these feature distributions might shift drastically in less formal contexts, such as social media or poetry. Finally, our extraction of deep syntactic features depends on the accuracy of existing NLP tools like *Stanza*. When applied to languages with limited training data, any errors in the underlying parser could potentially skew the linguistic analysis.

References

Anton Bakhtin, Sam Gross, Myle Ott, Yuntian Deng, Marc'Aurelio Ranzato, and Arthur Szlam. 2019. Real or fake? learning to discriminate machine from human generated text. *Preprint*, arXiv:1906.03351.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. *Preprint*, arXiv:1810.04805.

Ramesh Vaman Dhongde and Kashi Wali. 2009. *Marathi*, 3 edition, volume 13 of *London Oriental and African Languages Library*. John Benjamins, Amsterdam.

Sebastian Gehrmann, Hendrik Strobelt, and Alexander Rush. 2019. GLTR: Statistical detection and visualization of generated text. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. The llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Ganesh Jawahar, Muhammad Abdul-Mageed, and Laks V.S. Lakshmanan. 2020. Automatic detection of machine generated text: A critical survey. In *International Conference on Computational Linguistics*.

Emmanuel Joseph, Arthur Graham, and Matilda Ashford. 2025. Stylometric approaches for ai-text identification.

Kelly Kendro, Jeffrey Maloney, and Scott Jarvis. 2026. Do large language models produce texts with "human-like" lexical diversity? evidence from four chatgpt models. *International Journal of Applied Linguistics*.

Anoop Kunchukuttan. 2020. The IndicNLP Library. `https://github.com/anoopkunchukuttan/indic_nlp_library/blob/master/docs/indicnlp.pdf`.

S. Mittal, V. Magdum, S. Hiwarkhedkar, O. Dhekane, and R. Joshi. 2023. L3Cube-MahaNews: News-based short text and long document classification datasets in Marathi. In *International Conference on Speech and Language Technologies for Low-resource Languages*, pages 52–63. Springer.

Javier M. Moguerza and Alberto Muñoz. 2006. Support vector machines with applications. *Statistical Science*, 21(3).

Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D. Manning. 2020. Stanza: A python natural language processing toolkit for many human languages. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 101–108, Online. Association for Computational Linguistics.

Jonathon Shlens. 2014. A tutorial on principal component analysis. *Preprint*, arXiv:1404.1100.

A Extended Feature Loading Analysis

Our analysis grouped features into categories to evaluate overarching contributions to the principal components. The sum of squared loadings for the feature groups across PC1 and PC2 shows that Syntactic (Deep) structural features contribute the largest overall variance (0.5855), followed by Structural (0.4118) and Lexical features (0.3850). Predictability contributes a negligible amount (0.0150), reinforcing the necessity of morphologically-aware metric design in Marathi NLP tasks.

B Methodological Detail and Benchmarks

The Support Vector Machine (SVM) utilized an RBF kernel, trained using a standard 80-20 train-test split. The 2,700 human texts were precisely matched with 2,700 LLM-generated texts. Baseline BERT embeddings (from *bert-base-multilingual-cased*) were pooled and fed into an identical SVM pipeline to ensure a fair comparison. The poor performance of PCA on BERT

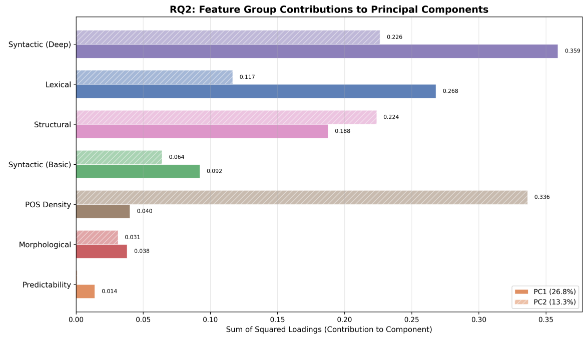

Figure 4: Contribution of each feature group to components. Deep Syntactic and Structural features drive the most variance.

(56.91% accuracy) implies that BERT's latent representations of Marathi do not linearly segregate semantic meaning from stylistic or generative traces within the first two principal components.